

Circles, Arcs & Sectors

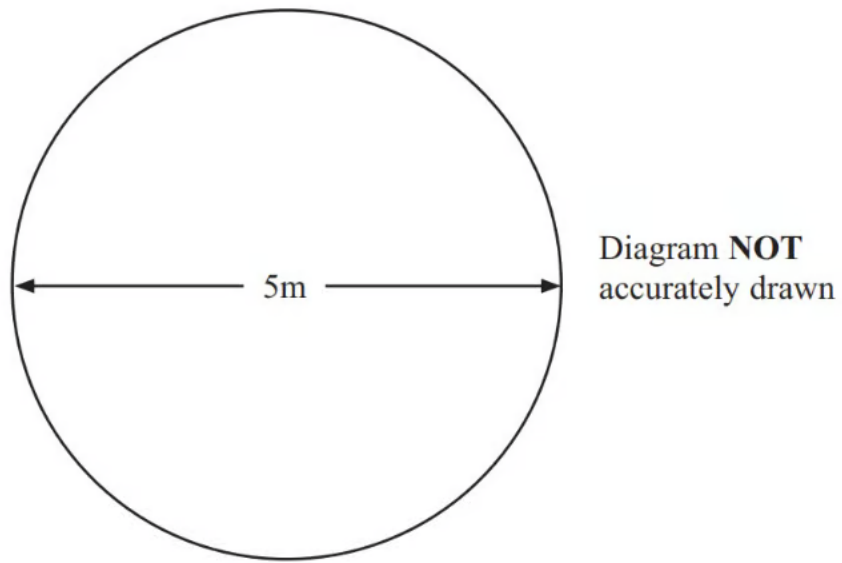
Question Paper

Course	Edexcel GCSE Maths
Section	4. Geometry & Measures
Topic	Circles, Arcs & Sectors
Difficulty	Medium

Time allowed: 80

Score: /60

Percentage: /100

Question 1

Jon has a flower garden in the shape of a circle.
The diameter of the garden is 5 metres.

Jon wants to put fencing around the edge of the garden.
The fencing costs £1.80 per metre.

Work out the total cost of the fencing.

£

[3 marks]

Question 2

The diagram shows a circle drawn inside a square.

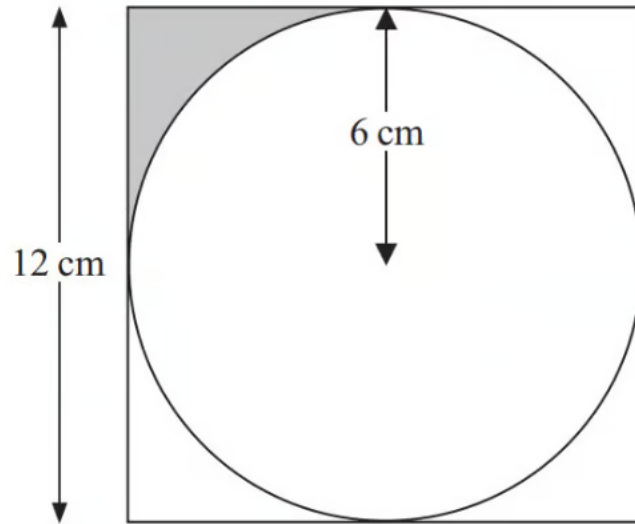


Diagram **NOT**
accurately drawn

The circle has a radius of 6 cm.

The square has a side of length 12 cm.

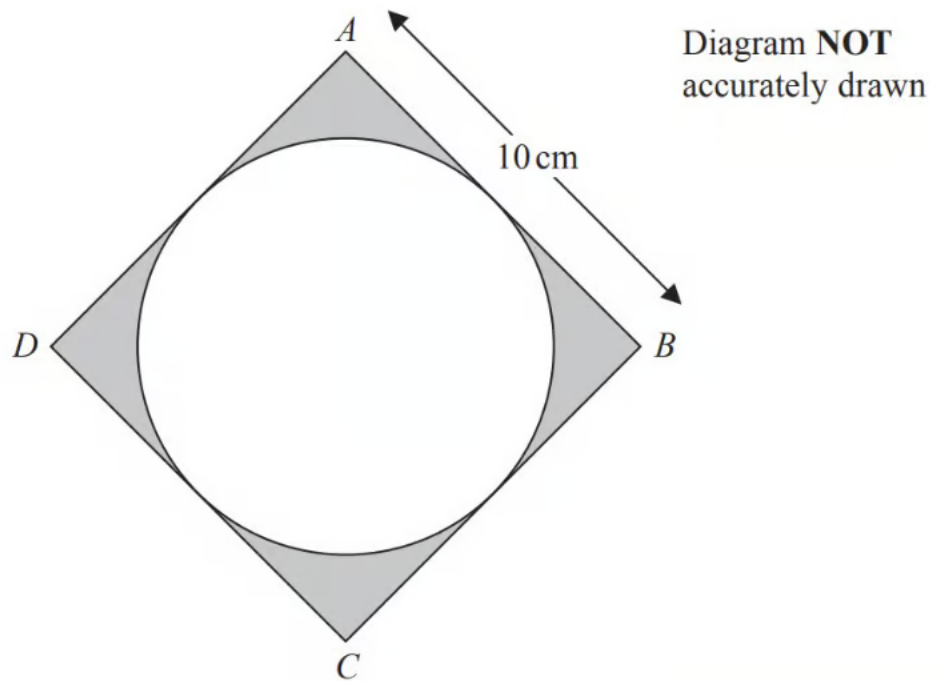
Work out the shaded area.

Give your answer in terms of π .

[3 marks]

Question 3

The diagram shows a circle inside a square.



$ABCD$ is a square of side 10 cm.

Each side of the square is a tangent to the circle.

Work out the total area of the shaded regions in terms of π .

Give your answer in its simplest form.

[3 marks]

Question 4

The diagram shows the top of Levi's birthday cake.

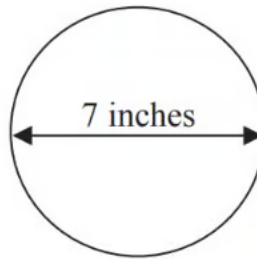


Diagram **NOT**
accurately drawn

The top of the cake is in the shape of a circle.

The diameter of the circle is 7 inches.

A ribbon is going to be put around the side of the cake.

Ribbons are sold in 50 cm lengths.

1 inch is 2.54 cm.

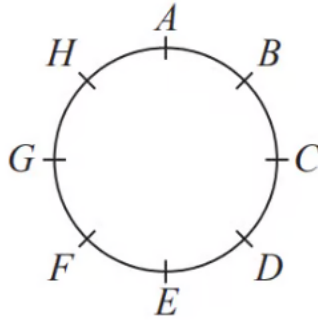
Work out if one length of ribbon is long enough to go all the way around the cake.

You must show your working.

[4 marks]

Question 5a

Hasmeet walks once round a circle with diameter 80 metres.



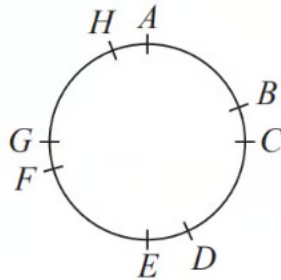
There are 8 points equally spaced on the circumference of the circle.

Find the distance Hasmeet walks between one point and the next point.

[2 marks]

Question 5b

Four of the points are moved, as shown in the diagram below.



Hasmeet walks once round the circle again.

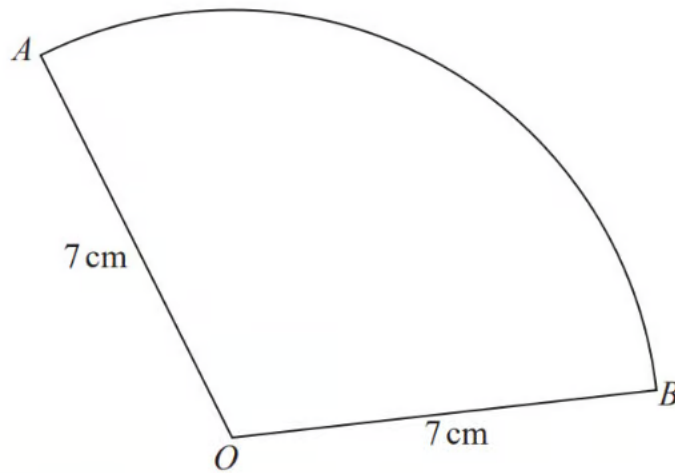
Has the mean distance that Hasmeet walks between one point and the next point changed?

You must give a reason for your answer.

[1 mark]

Question 6

OAB is a sector of a circle with centre O and radius 7 cm .



The area of the sector is 40 cm^2 .

Calculate the perimeter of the sector.

Give your answer correct to 3 significant figures.

[4 marks]

Question 7

The diagram shows a circle with centre O

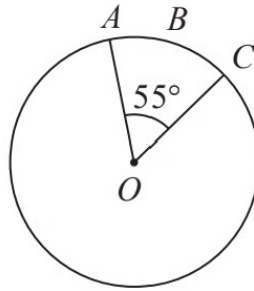


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

[4 marks]

Question 8

The region, shown shaded in the diagram, is a path.

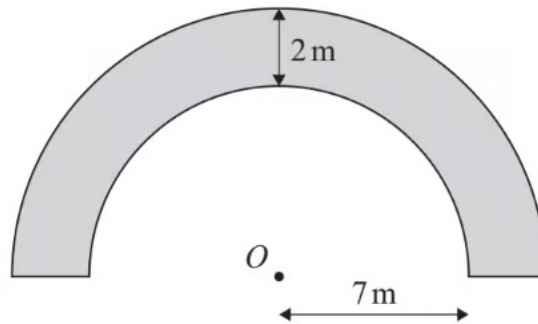


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

[3 marks]

Question 9

A circle centre O has radius 9 cm.

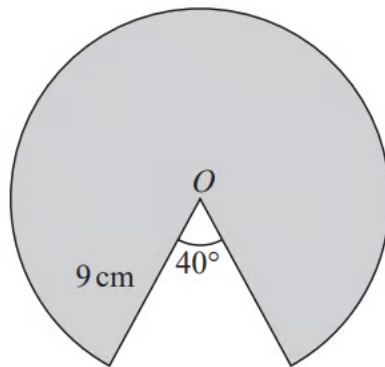


Diagram **NOT**
accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

[4 marks]

Question 10

The diagram shows Yuen's garden.

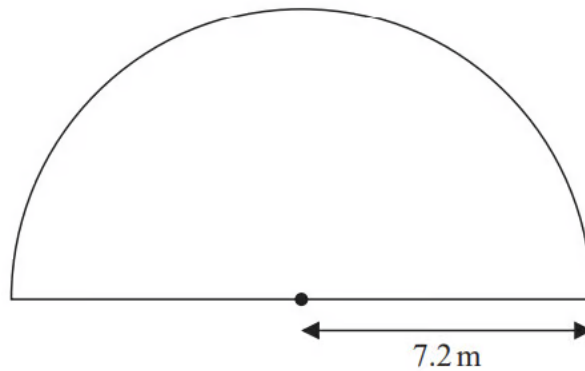


Diagram **NOT**
accurately drawn

The garden is in the shape of a semicircle of radius 7.2m .

Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.

Each box of grass seed contains enough seed to cover 6m^2 of the garden.

Has Yuen enough grass seed for his garden?

Show your working clearly.

[3 marks]

Question 11a

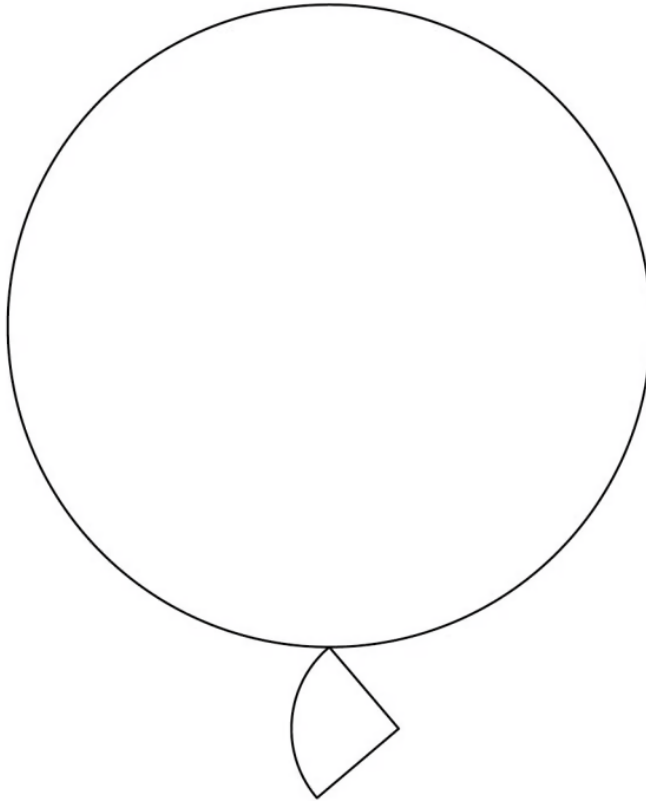
Two wire shapes make an earring.

The shapes are

a circle with radius 21 mm

and

a quarter circle.



Not drawn
accurately

radius of circle : radius of quarter circle = 7 : 2

Show that the radius of the quarter circle is 6 mm

[1 mark]

Question 11b

Work out the **total** length of the wire in the earring.

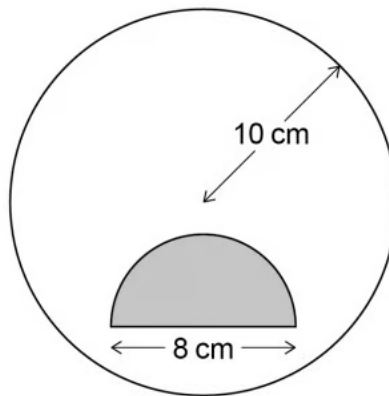
Give your answer in the form $a\pi + b$ where a and b are integers.

.....mm

[4 marks]

Question 12

A shaded semicircle is inside a circle as shown.



Not drawn
accurately

The **radius** of the circle is 10 cm

The **diameter** of the semicircle is 8 cm

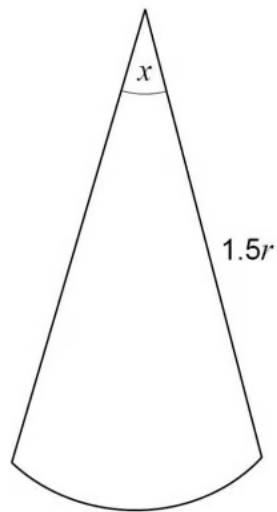
How many times bigger is the unshaded area than the shaded area?

[4 marks]

Question 13

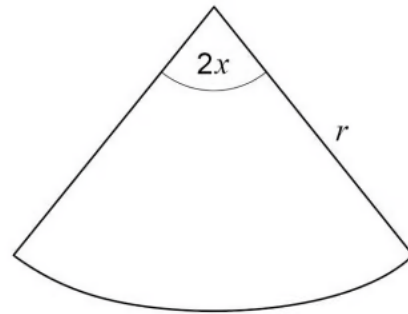
Here are two sectors from different circles.

Sector A



Sector B

Not drawn
accurately



Which sector has the bigger area?

Tick a box.

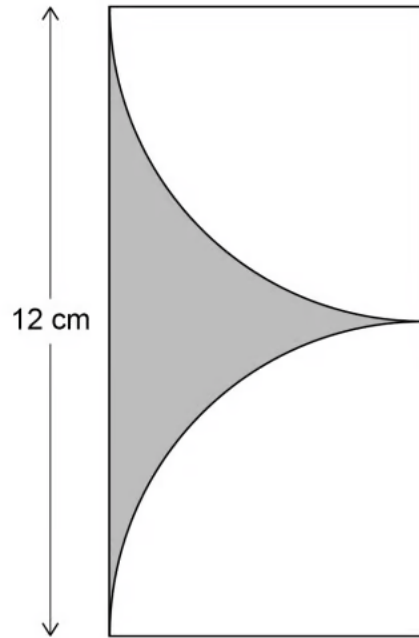
☐ Sector A ☐ Sector B

Show working to support your answer.

[2 marks]

Question 14

Two identical quarter circles are cut from a rectangle as shown.



Not drawn
accurately

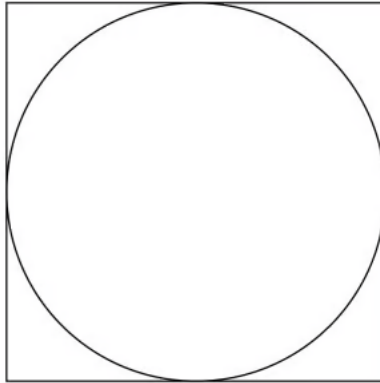
Work out the shaded area.

.....cm²

[4 marks]

Question 15

Here is a circle touching a square.



Not drawn
accurately

The area of the square is 64 cm^2

Work out the area of the circle.

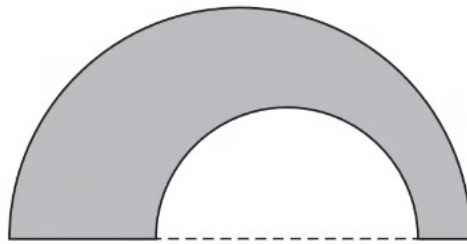
Give your answer in terms of π .

..... cm^2

[3 marks]

Question 16

The shape below is formed from two semicircles and a straight line.



Not to scale

The radius of the large semicircle is 8 cm.

The radius of the small semicircle is t cm.

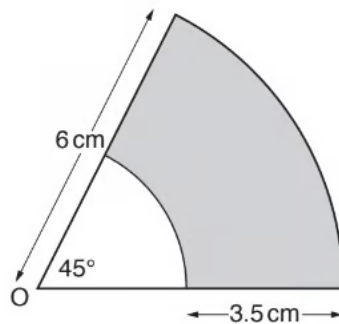
Find an expression, in terms of t , for the **exact perimeter** of the shaded shape.

..... cm

[3 marks]

Question 17

The design below is made from two sectors of circles, centre O.



Calculate the perimeter of the shaded part.

Give your answer correct to 3 significant figures.

..... cm

[5 marks]